VIVEK KRISHNA

18/12/2024

Code-E_(Phase-5)

Aakash
Medical | IIT-JEE | Foundations

Corporate Office : Aakash Tower, 8, Pusa Road, New Delhi-110005, ***Ph.***011-47623456

MM : 720 NCERT Booster Test Series for NEET (RM-P5)-2024-25_T01E **Time : 200 Min.**

Topics Covered:
Physics: Units and Measurements, Motion in a Straight Line, Motion in a Plane, Laws of Motion, Work, Energy and Power, System of Particles & Rotational Motion, Gravitation, Mechanical Properties of Solids, Mechanical Properties of Fluids
Chemistry: Some Basic Concepts of Chemistry, Structure of Atom, Classification of Elements and Periodicity in Properties, Chemical Bonding and Molecular Structure, Thermodynamics, Equilibrium, Redox Reactions, Electrochemistry
Botany: Cell: The Unit of Life, Cell Cycle and Cell Division, The Living World, Biological Classification, Morphology of Flowering Plants, Anatomy of Flowering Plants, Plant Kingdom
Zoology: Structural Organisation in Animals– Animal Tissues, Biomolecules, Breathing and Exchange of Gases, Body Fluids and Circulation, Excretory Products and their Elimination, Locomotion and Movement, Neural Control and Coordination

General Instructions :

1. There are two sections in each subject, i.e. **Section-A & Section-B**. You have to attempt all **35 questions** from Section-A & only **10 questions** from Section-B out of **15.**
2. Each question carries **+4 marks**. For every wrong response, **–1 mark** shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
3. Use blue/black ballpoint pen only to darken the appropriate circle.
4. Mark should be dark and completely fill the circle.
5. Dark only one circle for each entry.
6. Dark the circle in the space provided only.
7. Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

SECTION-A

1. A body is performing uniform circular motion. If the speed of the body gets halved whereas radius remains the same, then the ratio of initial and final centripetal accelerations is

(1) 1 : 2

(2) 2 : 1

(3) 1 : 4

(4) 4 : 1

2. The horizontal range of a projectile fired at an angle of 15° with the horizontal is 30 m. If it is fired at 45° to vertical with same speed its range will be

(1) 90 m

(2) 120 m

(3) 60 m

(4) 100 m

3. A particle is revolving in a circular path of radius 10 m with constant angular speed 20 rev/min. Then the angular acceleration of particle is

(1) π^2

(2) $3\pi^2$

(3) $5\pi^2$

(4) Zero

4. A boat is crossing a river, flowing with velocity 3 m/s. To reach exactly opposite side, boat takes 5 second. Width of the river is 20 m. The velocity of boat in still water should be

(1) 5 m/s

(2) 10 m/s

(3) 12 m/s

(4) 15 m/s

5. The potential energy of a long spring when stretched by 2 cm is U. If the spring is stretched by 8 cm, the potential energy will be

(1) 4 U

(2) 8 U

(3) 16 U

(4) $U/4$

6. If the impulse I of a body varies with time t as $I = (5t^2 - 10t + 1)$ kg m/s, then the change in momentum of body at $t = 1$ s is

(1) 3 kg m/s

(2) –4 kg m/s

(3) Zero

(4) –2 kg m/s

7. A machine gun fires 10 bullets per second with speed 10 m/s. If mass of each bullet is 300 g, then the force required to keep the gun stationary is

(1) 40 N

(2) 10 N

(3) 20 N

(4) 30 N

8. The equation of a projectile is, $y = \sqrt{3}x - \frac{gx^2}{2}$, the initial velocity is

(1) 4 ms^{-1}

(2) 2 ms^{-1}

(3) 1 ms^{-1}

(4) 5 ms^{-1}

9. Two soap bubbles of radii 3 cm and 4 cm are kept in touch with each other. What is radius of curvature of interface?

(1) 5 cm

(2) 7 cm

(3) 12 cm

(4) 2 cm

10. Work done by conservative forces acting on body is always equal to

(1) Negative of change in potential energy

(2) Magnitude of change in potential energy

(3) Change in kinetic energy

(4) Change in mechanical energy

11. Two balls, one of mass m and other of mass $2m$ are moving towards each other with respective speeds of 10 m/s and 20 m/s on a smooth horizontal floor. If after collision, they stick to each other and move as combined mass (perfectly inelastic collision), then the speed of combined mass will be

(1) 20 m/s

(2) 10 m/s

(3) 30 m/s

(4) 50 m/s

12. A ball (m = 1 kg) is dropped from height 20 m. If coefficient of a restitution in 0.9, the impulse imparted to the ground (by the ball) is

(1) 38 N s

(2) 40 N s

(3) 57 N s

(4) 81 N s

13. A projectile is projected with initial velocity $\left(10\hat{i} + 20\hat{j}\right)$ m/s from the ground. The velocity of the body just before hitting the ground (in m/s) is

(1) $-10\hat{i} - 20\hat{j}$

(2) $-10\hat{i} + 20\hat{j}$

(3) $10\hat{i} - 20\hat{j}$

(4) $10\hat{i} + 20\hat{j}$

14. The magnitude of average velocity is always equal to the average speed, when the particle moves

(1) On a curved path

(2) In the same direction

(3) With constant acceleration

(4) With constant retardation

15. The velocity vector of the motion of particle, described by the position vector $\vec{r} = 2t^2\hat{i} + t^3\hat{j}$ is

(1) $\vec{v} = 4t\hat{i} + 3t^2\hat{j}$

(2) $\vec{v} = 4\hat{i} + 6t\hat{j}$

(3) $\vec{v} = 2t^2\hat{i} + 3t^2\hat{j}$

(4) $\vec{v} = 2\hat{i} + \hat{j}$

16. The bulk modulus of an incompressible liquid is

(1) Zero

(2) Unity

(3) Infinity

(4) Between 0 to 1

17. Eight identical drops of water are falling in air, each with terminal velocity of 0.15 m/s. If they combine to form a single bigger drop, then terminal velocity of bigger drop will be

(1) 0.15 m/s

(2) 0.30 m/s

(3) 1.2 m/s

(4) 0.60 m/s

18. A body is projected at an angle 60° from surface of earth with velocity $\sqrt{2gR}$. Final K.E. of object at infinity is (symbols have usual meanings)

(1) Zero

(2) mgR

(3) Infinite

(4) $\frac{mgR}{2}$

19. A wheel rotates with a constant angular acceleration of 2.0 rad/s^2. If the wheel starts from rest, the number of revolutions it makes in first 10 seconds is approximately

(1) 8

(2) 16

(3) 24

(4) 32

20. A particle moves from $x = 0$ to $x = 5$ m, under the influence of a force F (in N) given by $F = 3x^2 - 2x + 7$. The work done by this force is

(1) 121 J

(2) 135 J

(3) 99 J

(4) 105 J

21. A disc of mass 'm' and radius 'r' is rotating with angular velocity ω_0. Another disc of mass $3m$ and radius $\frac{r}{3}$ is placed coaxially on the first disc gently. Final angular velocity of the combined system is

(1) ω_0

(2) $\frac{\omega_0}{2}$

(3) $2\omega_0$

(4) $\frac{3\omega_0}{4}$

22. A : The slope of stress-strain graph within proportionality limits is equal to modulus of elasticity.
R : Hooke's law is valid after elastic limit.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

23. When temperature of a liquid increases then its viscosity

(1) Increases

(2) Decreases

(3) Remains constant

(4) Depends on type of liquid

24. Liquid rises in capillary tube when angle of contact θ is

(1) $\frac{\pi}{2} < \theta < \pi$

(2) $0 < \theta < \frac{\pi}{2}$

(3) $\theta = \frac{\pi}{2}$

(4) $\theta = \pi$

25. A block of wood floats in water with $\left(\frac{2}{3}\right)^{rd}$ of its volume submerged. If the same block floats in an oil with $\left(\frac{1}{3}\right)^{rd}$ of its volume submerged, then density of oil (in kg/m^3) will be

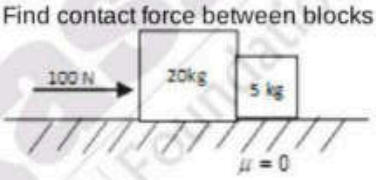
(1) 2

(2) $\frac{20}{3}$

(3) $\frac{200}{3}$

(4) 2000

26. Find contact force between blocks



(1) 20 N

(2) 40 N

(3) 60 N

(4) 50 N

27. A car of mass m is moving on a level circular track of radius R, if μ_s represents the static friction between the road and tyres of the car, the maximum speed of the car in circular motion is

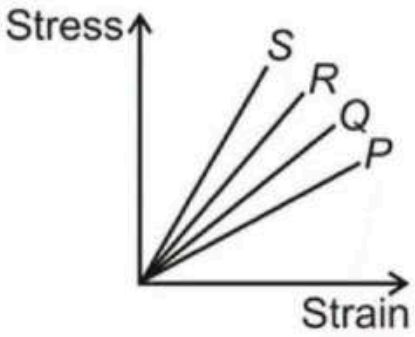
(1) $\sqrt{mRg/\mu_s}$

(2) $\sqrt{\mu_s Rg}$

(3) $\sqrt{\mu_s mRg}$

(4) $\sqrt{Rg/\mu_s}$

28. Stress-strain curve for four metals are shown in the figure. The minimum Young's modulus of elasticity is for metal



(1) *P*

(2) *Q*

(3) *R*

(4) S

29. **Statement A:** Work done by non conservative force is path independent.
Statement B: Energy is scalar quantity but power is a vector quantity.

(1) A and B are correct

(2) A and B are incorrect

(3) A is correct and B is incorrect

(4) A is incorrect and B is correct

30. Magnitude of vector product of $\vec{A}$ and $\vec{B}$ is $\sqrt{3}$ times than their scalar product. Then angle between vectors is

(1) $\frac{\pi}{4}$

(2) $\frac{\pi}{6}$

(3) $\frac{\pi}{3}$

(4) $\frac{\pi}{2}$

31. Two wires *A* and *B* are made up of same material, the ratio of their lengths and diameters are 1 : 2 and 2 : 1 respectively. If same pulling force is applied to each, then the ratio of their increase in lengths will be

(1) 2 : 1

(2) 1 : 4

(3) 1 : 8

(4) 16 : 1

32. A diver is 20 m below the surface of water. The pressure experienced by the diver is about

(1) 2×10^5 Pa

(2) 1×10^5 Pa

(3) 3×10^5 Pa

(4) Zero

33. A car moves with speed of 40 km/h for 2 hours and then with a speed of 60 km/h for 1 hour. The distance travelled by the car over entire motion is

(1) 36 km

(2) 140 km

(3) 180 km

(4) 20 km

34. The percentage error in mass and length of a cube is 2% and 1% respectively. The maximum percentage error in density will be

(1) 4%

(2) 5%

(3) 2%

(4) 1%

35. The speed of a projectile at its maximum height is $\frac{\sqrt{3}}{2}$ times of its initial speed. The angle of projection is

(1) 45°

(2) 30°

(3) 90°

(4) 60°

SECTION-B

36. In a river of width 200 m, flow speed is 10 ms^{-1}. A swimmer wants to cross the river along the shortest path & starts swimming with a speed of 20 ms^{-1}. Angle between his velocity and velocity of flow is

(1) 150°

(2) 90°

(3) 60°

(4) 120°

37. A lift moving vertically up, is experiencing decrease in speed at the rate 2 m/s^2. Normal force exerted on the floor of the lift by a person of mass 60 kg, standing in the lift is [Take g = 10 m/s^2]

(1) Equal to 600 N

(2) More than 600 N

(3) Less then 600 N

(4) Zero

38. A body of mass 2 kg is placed on smooth horizontal plane, pulled by the force of 2 N then the velocity of the body after 3 s is

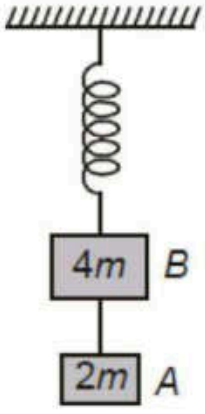
(1) 1 m/s

(2) 2 m/s

(3) 3 m/s

(4) 4 m/s

39. Two blocks *A* and *B* of masses 2*m* and 4*m* are connected by a string. The block of mass 4*m* is connected by a spring (massless). The string is suddenly cut. The ratio of magnitude of accelerations of masses 2*m* and 4*m* at that instant are



(1) 1 : 2

(2) 2 : 1

(3) 1 : 4

(4) 4 : 1

40. If force (F), work (W) and velocity (v) are taken as fundamental quantities, then the dimensional formula of time (T) is

(1) [W F V]

(2) [$W F^{-1} V^{-1}$]

(3) [$W^{-1} F V^{-2}$]

(4) [$W F^2 V^{-1}$]

41. The equation of path of a projectile is $y = 12x - \frac{3}{4}x^2$. (where x and y are in metre) What is the range of the projectile? $(g = 10\ m/s^2)$

(1) 36 m

(2) 20 m

(3) 16 m

(4) 90 m

42. A hiker stands on the edge of a cliff 122.5 m above the ground and throws a stone horizontally with an initial speed of 15 m s^{-1}. The vertical component of velocity of the stone when it hits the ground has magnitude nearly equal to

(1) 19.6 m s^{-1}

(2) 39.2 m s^{-1}

(3) 49 m s^{-1}

(4) 55 m s^{-1}

43. Kepler's second law is based on

(1) Newton's second law

(2) Special theory of relativity

(3) Stefan law

(4) Conservation of angular momentum

44. A wire of length *L* and cross-sectional area *A* is made of material of Young's Modulus (*Y*). The work done in stretching the wire by an amount *X* is

(1) $\frac{YAX^2}{2L}$

(2) $\frac{2YAX^2}{L}$

(3) $\frac{YAX^2}{L}$

(4) $\frac{4YAX^2}{L}$

45. At what altitude will the acceleration due to gravity be 25% of that at the earth's surface? (Radius of earth = *R*)

(1) $\frac{1}{4}R$

(2) R

(3) $\frac{3}{8}R$

(4) $\frac{R}{2}$

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46. A rope 2 cm in diameter breaks, if the tension in it exceeds 600 N. The maximum tension that a similar rope of diameter 3 cm can withstand is

(1) 600 N

(2) 900 N

(3) 450 N

(4) 1350 N

47. A spherical body falling through a viscous liquid of infinite extent finally attains a constant velocity, when

(1) Viscous drag – upthrust = weight

(2) Weight + viscous drag = upthrust

(3) Viscous drag + upthrust = weight

(4) Weight > upthrust + viscous drag

48. A : In every collision, linear momentum of the system is conserved.
R : In every collision, kinetic energy of the system is conserved.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

49. **Assertion (A) :** Pascal's law is the working principle of hydraulic brakes.
Reason (R) : Pressure = Thrust × Area.

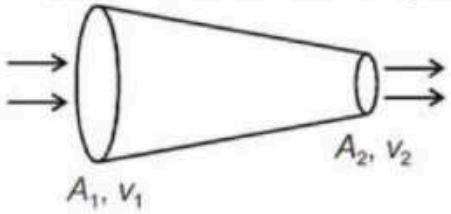
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(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

50. In the given figure if motion of liquid is streamline and $A_1 = 2A_2$, then ratio of speeds of liquid $\frac{v_1}{v_2}$ will be



(1) 1 : 2

(2) 1 : 4

(3) 4 : 1

(4) 2 : 1

CHEMISTRY

SECTION-A

51. Which of the following is the incorrect match?

	Prefix	Multiple
(1)	Milli	10^{-3}
(2)	Pico	10^{-12}
(3)	Nano	10^{-9}
(4)	Micro	10^{-4}

(1) (1)

(2) (2)

(3) (3)

(4) (4)

52. The number of significant figures in 0.001 is

(1) 1

(2) 2

(3) 3

(4) 4

53. Find number of oxygen atoms present in 100 mg of $CaCO_3$.
(Atomic mass of Ca = 40 u, C = 12 u, O = 16 u)

(1) 6.02×10^{23}

(2) 6.02×10^{20}

(3) 1.806×10^{21}

(4) 1.204×10^{20}

54. An element, X has the following isotopic composition : ^{92}X : 60% and ^{94}X : 40%; the weighed average atomic mass of the naturally occurring element X is

(1) 92.6 u

(2) 92.8 u

(3) 93.0 u

(4) 93.2 u

55. Match the following items in column I with column II and choose the correct option.

	Column I (quantum no. s)		Column II (orbital)
(a)	n = 2, l = 1	(i)	3*p*
(b)	n = 3, l = 2	(ii)	3*d*
(c)	n = 1, l = 0	(iii)	2*p*
(d)	n = 3, l = 1	(iv)	1*s*

(1) (a)(iii), (b)(i), (c)(iv), (d)(ii)

(2) (a)(iii), (b)(ii), (c)(iv), (d)(i)

(3) (a)(ii), (b)(i), (c)(iii), (d)(iv)

(4) (a)(i), (b)(ii), (c)(iii), (d)(iv)

56. The number of protons, neutrons and electrons in $^{40}_{20}Ca^{2+}$ ion respectively are

(1) 20, 20, 20

(2) 18, 20, 20

(3) 20, 20, 18

(4) 20, 18, 20

57. A golf ball has a mass of 50 g, and a speed of 30 m/s. If the speed can be measured within accuracy of 1%, then the uncertainty in the position will be

(1) 3.5×10^{-33} m

(2) 7.5×10^{-33} cm

(3) 2×10^{-33} m

(4) 1.39×10^{33} km

58. A 200 watt bulb emits monochromatic light of wavelength 800 Å. The approximate number of photons emitted per second by the bulb is ($h = 6.6 \times 10^{-34}$ Js)

(1) $6 \times 10^{19} s^{-1}$

(2) $7 \times 10^{19} s^{-1}$

(3) $9 \times 10^{19} s^{-1}$

(4) $8 \times 10^{19} s^{-1}$

59. Splitting of spectral lines under the influence of magnetic field is called

(1) Photoelectric effect

(2) Stark effect

(3) Zeeman effect

(4) Diffraction

60. The ionic radii of N^{3-}, O^{2-}, F^-, Na^+ follow the order

(1) $N^{3-} > O^{2-} > F^- > Na^+$

(2) $N^{3-} > Na^+ > O^{2-} > F^-$

(3) $Na^+ > O^{2-} > N^{3-} > F^-$

(4) $Na^+ > F^- > O^{2-} > N^{3-}$

61. Given below are two statements one is **Assertion (A)** other is **Reason (R)**.
Assertion (A): First ionization energy for nitrogen is higher than that of oxygen.
Reason (R): Across a period effective nuclear charge decreases.
In the light of above statements choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of the (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of the (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

62. Eka-silicon is also called as

(1) Gallium

(2) Aluminium

(3) Indium

(4) Germanium

63. Element of highest electronegativity among the following is

(1) O

(2) S

(3) Se

(4) Te

64. Consider the following statements about resonance
(i) The canonical forms have no real existence
(ii) There is equilibrium between the canonical forms
(iii) Resonance averages the bond characteristics as a whole.
Choose the correct statement.

(1) Only (i)

(2) Only (i) & (ii)

(3) Only (ii) & (iii)

(4) Only (i) & (iii)

65. Consider the following two statements.
Statement I: Magnitude of H-bonding depends on the physical state of the compound.
Statement II: Magnitude of H-bonding is maximum in liquid state and minimum in solid state.

In the light of above statements, choose the correct option.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

66. In the process $O_2 \longrightarrow O_2^+ + e^-$
The electron lost is from

(1) Bonding σ-orbital

(2) Antibonding σ-orbital

(3) Bonding π-orbital

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67. The formal charge present on central atom of NH_4^+ ion is

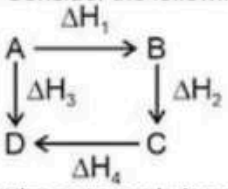
(1) Zero

(2) +1

(3) –1

(4) +2

68. Consider the following reaction path



The correct relation of enthalpies is

(1) $\Delta H_1 + \Delta H_3 = \Delta H_2 - \Delta H_4$

(2) $\Delta H_1 + \Delta H_2 = \Delta H_3 - \Delta H_4$

(3) $\Delta H_1 + \Delta H_4 = \Delta H_2 + \Delta H_3$

(4) $\Delta H_1 + \Delta H_4 = 2\Delta H_2 - 3\Delta H_3$

69. Which of the following is extensive property?

(1) Molar heat capacity

(2) Vapour pressure

(3) Gibb's free energy

(4) Surface tension

70. According to third law of thermodynamics which property of any pure crystalline substances approaches zero as the temperature approaches absolute zero?

(1) U

(2) H

(3) G

(4) S

71. If 70 J heat is supplied to a system, and gas has done 150 J expansion work during a thermodynamic process, then change in internal energy of the system will be

(1) +60 J

(2) –80 J

(3) –60 J

(4) +80 J

72. The species having bond order same as that of CN^- are
(a) CO (b) O_2
(c) H_2 (d) N_2

(1) (a) & (b)

(2) (a) & (d)

(3) (c) & (d)

(4) (a), (b) & (d)

73. The equilibrium constant K_C for the reaction is 4×10^2 at 300 K. The $\Delta G^\ominus$ of the reaction is

(1) + 6.48 kJ mol^{-1}

(2) + 14.96 kJ mol^{-1}

(3) – 6.48 kJ mol^{-1}

(4) – 14.96 kJ mol^{-1}

74. Given below are two statements one is assertion (A) other is reason (R).
Assertion (A) : The ionisation of CH_3COOH is suppressed on adding CH_3COONa.
Reason (R) : Common ion effect is observed in the mixture of CH_3COOH and CH_3COONa.
In the light of above statements choose the correct option.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

75. The pH of 10^{-8} M NaOH solution is

(1) 6.98

(2) 6

(3) 7.04

(4) 8

76. Standard molar entropy of X_2, Y_2 and XY_3 are 60, 40 and 50 J K^{-1} mol^{-1} respectively.
For the reaction, $\frac{1}{2}X_2 + \frac{3}{2}Y_2 \rightarrow XY_3$, ΔH = –30 kJ. To be at equilibrium, the temperature will be

(1) 1000 K

(2) 1250 K

(3) 450 K

(4) 750 K

77. All of the following are heterogeneous equilibria except

(1) $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

(2) $Ni(s) + 4CO(g) \rightleftharpoons Ni(CO)_4(g)$

(3) $Ag_2O(s) + 2HNO_3(aq) \rightleftharpoons 2AgNO_3(aq) + H_2O(l)$

(4) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

78. The reaction that is the example of redox decomposition reaction is

(1) $3Mg + N_2 \longrightarrow Mg_3N_2$

(2) $2KClO_3 \xrightarrow{\Delta} 2KCl + 3O_2$

(3) $2F_2 + 2OH^-(aq.) \rightarrow 2F^- + OF_2 + H_2O$

(4) $2Al + 6HCl \longrightarrow 2AlCl_3 + 3H_2$

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79. Oxidation state of middle carbon of C_3O_2 molecule is

(1) 0

(2) –2

(3) +2

(4) –4

80. For the redox reaction,

$Cr_2O_7^{2-} + SO_3^{2-} + H^+ \rightarrow Cr^{3+} + SO_4^{2-} + H_2O$

The correct coefficients of the products for the balanced equation are

	Cr^{3+}	SO_4^{2-}	H_2O
(1)	3	2	4
(2)	2	3	4
(3)	4	2	3
(4)	2	4	3

(1) (1)

(2) (2)

(3) (3)

(4) (4)

81. Oxidation number of Cl in $HClO_4$ is

(1) + 1

(2) + 3

(3) + 5

(4) + 7

82. In lead storage battery the electrolyte used is

(1) Paste of KOH and ZnO

(2) 38% solution of H_2SO_4

(3) Aqueous $ZnSO_4$

(4) Moist paste of NH_4Cl

83. Given $E^\circ_{Zn^{2+}/Zn} = -0.76$ V, $E^\circ_{Fe^{2+}/Fe} = -0.44$ V

$E^\circ_{Ni^{2+}/Ni} = -0.25$ V and

$E^\circ_{Cu^{2+}/Cu} = 0.34$ V

Select the correct statement among the following

(1) Zn^{2+} can oxidise all Fe, Ni and Cu

(2) Fe^{2+} can oxidise Ni and Cu but not Zn

(3) Zn can reduce Fe^{2+} and Ni^{2+} but not Cu^{2+}

(4) Fe can reduce Ni^{2+} and Cu^{2+} but not Zn^{2+}

84. The electrode potentials for $Fe^{3+} + 3e^- \rightarrow Fe$ and $Fe^{2+} + 2e^- \rightarrow Fe$ are –0.04 V and –0.44 V respectively. The value of $E^\circ_{Fe^{3+}/Fe^{2+}}$ will be

(1) 0.41 V

(2) 0.52 V

(3) 1.00 V

(4) 0.76 V

85. Given below are two statements.

Statement (I) : Reactions with positive cell potential are spontaneous

Statement (II) : For an electrochemical cell $\Delta G = -nFE$

In the light of above statements, choose the **correct** answer.

(1) Both statement I and statement II are true

(2) Both statement I and statement II are false

(3) Statement I is correct but statement II is false

(4) Statement I is incorrect but statement II is true

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SECTION-B

86. An organic compound contains carbon, hydrogen and oxygen. Its elemental analysis gave C = 40% and H = 6.67%. The empirical formula of the compound would be

(1) HCO

(2) H_2CO

(3) HC_2O

(4) H_2C_2O

87. Correct scientific notation for 232.508 is

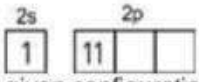
(1) 2.32508×10^2

(2) 2.32508×10^{-2}

(3) 0.232508×10^3

(4) 0.232508×10^{-3}

88. 2s 2p



given configuration violates

(1) Pauli's and Aufbau only

(2) Pauli's and Hund's only

(3) Aufbau and Hund's only

(4) Aufbau, Pauli's and Hund's

89. Consider the following statements for photoelectric effect.
(a) The number of electrons ejected is proportional to the intensity or brightness of light.
(b) There is a time lag between the striking of light beam and the ejection of electrons from the metal surface.
(c) For each metal there is a characteristic maximum frequency known as threshold frequency above which photoelectric effect is not observed.
The incorrect statement(s) is/are

(1) (a) only

(2) (a) & (b) only

(3) (b) & (c) only

(4) (a), (b) & (c)

90. Given below are two statements: One is labelled as **Assertion (A)** and other is labelled as **Reason (R)**.
Assertion (A) : Electron affinity of chlorine is higher than fluorine.
Reason (R) : Electronegativity of F is higher than chlorine.
In the light of above statements choose the correct option.

(1) Both (A) and (R) are true and the (R) is the correct explanation of the (A)

(2) Both (A) and (R) are true but the (R) is not the correct explanation of the (A)

(3) (A) is true statement but (R) is false

(4) Both (A) and (R) are false statements

91. Bond length of H-X is 0.94 Å and observed dipole moment of H-X is 1.8 D, then % of ionic character of H-X is

(1) 21.2%

(2) 62.3 %

(3) 39.9%

(4) 78.4%

92. Match list I with list II.

	List I (Compound)		List II (Hybridisation)
a.	XeF_4	(i)	sp^2
b.	SO_3	(ii)	sp^3d^2
c.	SO_4^{2-}	(iii)	sp^3d
d.	XeF_2	(iv)	sp^3

The correct match is

(1) a(iv), b(i), c(ii), d(iii)

(2) a(ii), b(i), c(iii), d(iv)

(3) a(iii), b(i), c(iv), d(ii)

(4) a(ii), b(i), c(iv), d(iii)

93. If the enthalpy of formation of $N_2O(g)$, $N_2O_4(g)$, $CO(g)$ and $CO_2(g)$ is 800, 15, –100, and –400 kJ/mol respectively, then Δ_rH for the given reaction is
$N_2O_4(g) + 3\,CO(g) \rightarrow N_2O(g) + 3\,CO_2(g)$

(1) +90 kJ/mol

(2) –115 kJ/mol

(3) +115 kJ/mol

(4) –500 kJ/mol

94. Which among the following match is incorrect?

	$\Delta_rH°$	$\Delta_rS°$	$\Delta_rG°$	Description
(1)	–	+	–	Reaction spontaneous at all temperature
(2)	–	–	–	Reaction spontaneous at low temperature
(3)	+	+	–	Reaction nonspontaneous at high temperature
(4)	+	–	+	Reaction nonspontaneous at all temperature

(1) (1)

(2) (2)

(3) (3)

(4) (4)

95. If solubility of sparingly soluble salt A_2X_3 is s $molL^{-1}$ then its K_{sp} is

(1) s^2

(2) $4s^3$

(3) $9s^4$

(4) $108s^5$

96. Which pair can act as acidic buffer?

(1) HCN and KCN

(2) NH_4OH and NH_4Cl

(3) $HClO_4$ and $KClO_4$

(4) H_2SO_4 and $NaHSO_4$

97. pH of a solution of 0.1 M $CH_3COONH_4(aq)$ is (given: $K_a(CH_3COOH) = K_b(NH_4OH) = 1.8 \times 10^{-5}$)

(1) 1

(2) 9

(3) 7

(4) 13

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98. Given below are two statements: One is labelled as **Assertion (A)** and other is labelled as **Reason (R)**.

Assertion (A): Λ_m° for weak electrolyte cannot be obtained by extrapolation of Λ_m to zero concentration.

Reason (R): Λ_m° for weak electrolytes is obtained by using Kohlrausch law of independent migration of ions.
In the light of above statements choose the correct option.

(1) Both (A) and (R) are true and the (R) is the correct explanation of the (A)

(2) Both (A) and (R) are true but the (R) is not the correct explanation of the (A)

(3) (A) is true statement but (R) is false

(4) Both (A) and (R) are false statements

99. Mass of silver deposited at cathode by passage of 9.65 A current for 1000 sec is (Ag = 108 u)

(1) 27 g

(2) 54 g

(3) 10.8 g

(4) 216 g

100. Role of SO_2 in the following reactions respectively are

(a) $Fe_2(SO_4)_3 + SO_2 + 2H_2O \rightarrow 2FeSO_4 + 2H_2SO_4$

(b) $2H_2S + SO_2 \rightarrow 2H_2O + 3S$

(1) Reducing in (a) and oxidising in (b)

(2) Oxidising in (a) and reducing in (b)

(3) Oxidising in both (a) and (b)

(4) Reducing in both (a) and (b)

BOTANY

SECTION-A

101. Match the following columns and select the **correct** option.

Column I		Column II
a. Chloroplast	(i)	Cristae
b. Plasmodesmata	(ii)	Thylakoid
c. Mitochondria	(iii)	Calcium pectate
d. Middle lamella	(iv)	Connects cytoplasm of adjacent cells

(1) a(ii), b(iv), c(i), d(iii)

(2) a(ii), b(i), c(iii), d(iv)

(3) a(iii), b(iv), c(ii), d(i)

(4) a(iv), b(i), c(ii), d(iii)

102. A polysome contains

(1) Many ribosomes and a single mRNA

(2) Single ribosome and many mRNAs

(3) Single ribosome and single mRNA

(4) Many ribosomes and many mRNAs

103. Read the following statements and select the **correct** option.

Statement A : Neutral solutes may move across plasma membrane by the process of simple diffusion along the concentration gradient.

Statement B : The quasi-fluid nature of proteins, enables their lateral movement within the overall bilayer.

(1) Only statement A is incorrect

(2) Only statement B is incorrect

(3) Both statements A and B are correct

(4) Both statements A and B are incorrect

104. Choose the feature which is common in mitochondria, chloroplast and prokaryotic cell?

(1) Single stranded DNA

(2) Presence of 70S ribosomes

(3) Absence of enzymes

(4) Surrounded by rigid cell wall

105. Match the following columns and select the **correct** option.

	Column I		Column II
a.	Anton Von Leeuwenhoek	(i)	Fluid mosaic model of plasma membrane
b.	Rudolf Virchow	(ii)	First described nucleus as a cell organelle
c.	Robert Brown	(iii)	First saw and described a live cell
d.	Singer and Nicolson	(iv)	*Omnis cellula-e cellula*

(1) a(iii), b(iv), c(ii), d(i)

(2) a(ii), b(iv), c(iii), d(i)

(3) a(iii), b(i), c(iv), d(ii)

(4) a(iv), b(ii), c(i), d(iii)

106. Nuclear envelope in a nucleus

(1) Separates nucleoplasm from the cytoplasm

(2) Surround a perinuclear space of 10 – 100 µm in width

(3) Has ribosomes only on the nucleoplasmic surface

(4) Contains nuclear pore complex that controls one way (only towards outside) movement of substances

107. Kinetochores are

(1) Terminal ends of chromosomes

(2) Present around secondary constriction and help in crossing over

(3) Present near nuclear envelope and take part in spindle formation

(4) Present around centromere and provide a site for attachment to the spindle fibers

108. There are five different stages of prophase-I i.e., Leptotene (L), Pachytene (P), Diplotene (D), Zygotene (Z) and Diakinesis (D_i).

Arrange the stages in **correct** sequence.

(1) $D \rightarrow P \rightarrow L \rightarrow D_i \rightarrow Z$

(2) $L \rightarrow Z \rightarrow P \rightarrow D \rightarrow D_i$

(3) $Z \rightarrow L \rightarrow D_i \rightarrow P \rightarrow D$

(4) $P \rightarrow L \rightarrow Z \rightarrow D_i \rightarrow D$

109. In equational division of a cell, how many stages are observed in nuclear division?

(1) One

(2) Three

(3) Two

(4) Four

110. During the movement of chromosome away from the equatorial plate the __A__ of each chromosome is towards the __B__.

Select the option that **correctly** fills the blanks A and B.

	A	B
(1)	Arms	Poles
(2)	Chromatids	Poles
(3)	Centromere	Poles
(4)	Centromere	Equator

(1) (1)

(2) (2)

(3) (3)

(4) (4)

111. Crossing over is exchange of genetic material between

(1) Sister chromatids of homologous chromosomes

(2) Non-sister chromatids of non-homologous chromosomes

(3) Non-sister chromatids of homologous chromosomes

(4) Sister chromatids of non-homologous chromosomes

112. Interkinesis is a metabolic stage between __a__ and __b__.

	(a)	(b)
(1)	Telophase II	Prophase I
(2)	Telophase II	Prophase II
(3)	Telophase I	Prophase I
(4)	Telophase I	Prophase II

(1) (1)

(2) (2)

(3) (3)

(4) (4)

113. At which of the given stages of prophase-I of meiosis, nucleolus disappears and nuclear envelope disintegrates?

(1) Zygotene

(2) Pachytene

(3) Diplotene

(4) Diakinesis

114. Prophase I is similar to prophase II as

(1) Both always occur in haploid cells

(2) Both include five substages

(3) Nuclear membrane disappears in both

(4) Both are equally complex and prolonged stages

115. The taxon that are equivalent to Diptera and Insecta respectively are

(1) Anacardiaceae and Sapindales

(2) Poales and Angiospermae

(3) Sapindales and Dicotyledonae

(4) Monocotyledonae and Angiospermae

116. Read the following statements and select the **correct** option.
Statement-A : Cellular organisation of the body is the defining feature of all life forms.
Statement-B : Isolated metabolic reactions *in-vitro* are not living reactions.

(1) Both statements are correct

(2) Only statement A is correct

(3) Only statement B is correct

(4) Both statements are incorrect

117. Match the following columns and select the **correct** option.

	Column I		Column II
a.	*Penicillium*	(i)	Zygospore
b.	*Rhizopus*	(ii)	Imperfect fungus
c.	*Ustilago*	(iii)	Ascospore
d.	*Colletotrichum*	(iv)	Basidiospore

(1) a(iii), b(i), c(iv), d(ii)

(2) a(i), b(iii), c(iv), d(ii)

(3) a(iii), b(i), c(ii), d(iv)

(4) a(ii), b(i), c(iv), d(iii)

118. Amoeboid protozoans differ from all the other protozoan because they

(1) Use cilia for locomotion

(2) Have two types of nuclei

(3) Use flagella for movement

(4) Use pseudopodia for capturing prey

119. Read the following statements and choose the **correct** option.
Statement A: Basidiospores are the exogenously produced sexual spores.
Statement B: Basidiospores are produced on sex organs.

(1) Both the statements are correct

(2) Both the statements are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

120. Mark True (T) or False (F) for the statements given below.
I. Only the sexual phases of imperfect fungi are known.
II. The mycelium is septate and branched in Deuteromycetes.
III. Members of Deuteromycetes help in mineral cycling.

	I	II	III
(1)	T	T	F
(2)	F	T	T
(3)	F	F	T
(4)	T	F	F

(1) (1)

(2) (2)

(3) (3)

(4) (4)

121. Dinoflagellates differ from euglenoids, as the former

(1) Can predate on other smaller organisms

(2) Have majority of members that are fresh water organisms

(3) Have two flagella

(4) Have cell wall with stiff cellulose plates on the outer surface

122. Pulvinus is

(1) Swollen leaf base

(2) Cylindrical stalk that joins the leaf base with lamina

(3) Lateral small leaf like structure

(4) Modified petiole

123. In which of the following plants,
the floral formula has $\overset{\frown}{P_{(3+3)}}\,A_{3+3}$ symbol?

(1) *Petunia*

(2) Chilli

(3) Tulip

(4) Lupin

124. In racemose inflorescence

(1) Flowers are borne in basipetal order

(2) Main axis continues to grow

(3) Growth of main axis is limited

(4) Younger flower are present at the base and older flowers are present towards the apex

125. A tendril in cucumber is

(1) A modified leaf

(2) A modified axillary bud

(3) Present at internodes

(4) Woody and pointed

126. Arrangement of veins and veinlets in leaf lamina is called

(1) Venation

(2) Phyllotaxy

(3) Aestivation

(4) Placentation

127. Mark the **mismatched** pair.

	Union of stamens		Plants
(1)	Epipetalous	–	Brinjal
(2)	Polyadelphous	–	*Citrus*
(3)	Monadelphous	–	Pea
(4)	Epiphyllous	–	Lily

(1) (1)

(2) (2)

(3) (3)

(4) (4)

128. "Protoxylem lies towards the $\underline{A}$ in stem. Such arrangement of xylem is called $\underline{B}$." Select the **correct** option for A and B.

(1) **A** – Centre, **B** – Exarch

(2) **A** – Periphery, **B** – Exarch

(3) **A** – Centre, **B** – Endarch

(4) **A** – Periphery, **B** – Endarch

129. Which is **not** true about stomata?

(1) Present in epidermis of leaves

(2) Regulate the process of transpiration

(3) Composed of guard cells

(4) In grasses the guard cells are bean shaped

130. Casparian strips
(a) Are found on endodermis
(b) Are made of impermeable waxy material
(c) May be seen in roots and stems
(d) Allow passage of water molecules through them
Choose the **correct** one.

(1) Only a and b

(2) Only b, c and d

(3) Only a, b and c

(4) Only b and c

131. Among the following anatomical features, how many is/are **correct** for dicot stem?
a. Pericycle as semi-lunar patches of sclerenchyma.
b. Endodermis is referred to as the starch sheath.
c. The phloem parenchyma is absent.
d. Collenchymatous hypodermis.

(1) Four

(2) Three

(3) Two

(4) One

132. Select the structure **not** found in isobilateral leaf.

(1) Bulliform cells

(2) Palisade mesophyll

(3) Conjoint vascular bundle

(4) Stoma on upper surface

133. Which of the following plants belongs to the class Lycopsida?

(1) *Lycopodium*

(2) *Equisetum*

(3) *Pteris*

(4) *Adiantum*

134. In class Phaeophyceae, the plant body is usually attached to the substratum by a ___A___ and has a stalk, the ___B___ and leaf like photosynthetic part, the ___C___.
Choose the **correct** option for A, B and C.

	A	B	C
(1)	Holdfast	Stipe	Frond
(2)	Stipe	Holdfast	Frond
(3)	Frond	Stipe	Holdfast
(4)	Stipe	Frond	Holdfast

(1) (1)

(2) (2)

(3) (3)

(4) (4)

135. Read the following statements and select the **correct** option
Assertion (A): In gymnosperms, the male and the female gametophytes are said to have no independent free living existence.
Reason (R): In gymnosperms, the male and female gametophytes remain within the sporangia retained on the sporophytes.

(1) Both (A) and (R) are correct but (R) is not correct explanation of (A)

(2) (A) is correct but (R) is incorrect

(3) Both (A) and (R) are correct and (R) is correct explanation of (A)

(4) (A) is incorrect but (R) is correct

SECTION-B

136. The core of eukaryotic cilia and flagella is called axoneme which is made up of

(1) Filament, hook and basal body

(2) 9 centrally located doublets and 2 microtubules at periphery

(3) 2 centrally located microtubules and 9 doublets at periphery

(4) Central hub and 11 peripheral doublets

137. Endomembrane system in eukaryotic cells includes

(1) Both single and double membrane bound cell organelles.

(2) Endoplasmic reticulum, Golgi complex, lysosome and vacuole.

(3) Non-coordinated cell organelles.

(4) Chloroplasts, Endoplasmic reticulum and vacuole in plant cells.

138. Thick and tough glycocalyx found in some bacteria is called

(1) Slime layer

(2) Mesosome

(3) Capsule

(4) Lamellae

139. M-phase starts with

(1) Condensation of chromatin materials

(2) Duplication of DNA

(3) Division of cytoplasm

(4) Metabolic preparation of cell for its division

140. Arrange the following events of meiosis in their **correct** sequence of occurrence and choose the **correct** option.
a. Chromosomal synapsis
b. Dissolution of synaptonemal complex
c. Alignment of univalents at equator
d. Terminalisation of chiasmata

(1) a → b → d → c

(2) a → d → b → c

(3) b → a → c → d

(4) c → b → a → d

141. What will be the amount of DNA in each meiosis II product if the mother cell has 40 pg of DNA in G_2 phase?

(1) 10 pg

(2) 20 pg

(3) 40 pg

(4) 80 pg

142. ICBN stands for

(1) International Code for Bacterial Nomenclature

(2) Indian Code for Botanical Nomenclature

(3) International Code for Biological Nomenclature

(4) International Code for Botanical Nomenclature

143. Viroids

(1) Are infectious proteins

(2) Have capsid as outer covering

(3) Were discovered by T.O. Diener

(4) Are viruses with ssDNA

144. Slime moulds

(1) Are autotrophs

(2) Produce spores under unfavourable conditions

(3) Lack cell wall throughout the life cycle

(4) Are unicellular prokaryotes

145. How many of the given features are related to plant family to which Indigofera belongs?
(a) Diadelphous stamens
(b) Bicarpellary ovary
(c) Vexillary aestivation
(d) Polypetalous corolla
(e) Non endospermic seeds

(1) 5

(2) 4

(3) 2

(4) 3

146. A plant 'X' was studied by a student. He noted some features of the plant such as
(a) In flowers, petals have valvate aestivation
(b) Stamens are epipetalous
(c) Leaves are arranged in alternate fashion
Identify the plant 'X' on the basis of above given features.

(1) Mustard

(2) China rose

(3) Bean

(4) Tomato

147. Arrange the following components in an order from inside to outside, in a dicot stem, then choose **correct** option.
A. Vascular bundles
B. Pith
C. Pericycle
D. Cortex

(1) A – B – C – D

(2) D – C – A – B

(3) B – A – C – D

(4) B – D – C – A

148. The figure depicted below is transverse section of



(1) Monocot stem

(2) Monocot root

(3) Dicot stem

(4) Dicot root

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149. Kelps are

(1) Filamentous red algae

(2) Profusely branched brown algae

(3) Profusely branched red algae

(4) Colonial green algae

150. Pteridophytes differ from bryophytes in

(1) Lacking root, stem and leaves

(2) Being non-photosynthetic

(3) Having vascular tissues

(4) Requiring water for fertilisation

ZOOLOGY

SECTION-A

151. Organic compound found in living system that can produce zwitterionic form is

(1) Glucose

(2) Valine

(3) Lecithin

(4) Adenine

152. The type of epithelium present in the inner lining of ducts of salivary glands is also found in the

(1) Blood vessels

(2) Pharynx

(3) Small intestine

(4) PCT of nephron

153. Choose the correct option to complete the analogy.
Trypsin : Enzyme :: Insulin : _____

(1) Fat

(2) Hormone

(3) Nucleic acid

(4) Nucleotide

154. Which of the following is a set of aromatic amino acids?

(1) Valine, Phenylalanine

(2) Tyrosine, Tryptophan

(3) Glutamic acid, Phenylalanine

(4) Glycine, Serine

155. Rate of reaction double or halves with every _____ °C change in either direction. Choose the option that fills the blank **correctly**.

(1) 20

(2) 15

(3) 10

(4) 25

156. _____ is a co-factor for the proteolytic enzyme carboxypeptidase.
Select the correct option to fill in the blank.

(1) Magnesium

(2) Zinc

(3) Haem

(4) NADP

157. How many of the structures given in the box below are examples of connective tissue?

Cartilage, Bone, Tendon, Ligament, Neuroglia, Blood

Choose the **correct** option.

(1) Three

(2) Four

(3) Five

(4) Six

158. Which of the following statements is **incorrect** w.r.t. the simple columnar epithelium?

(1) It is composed of single layer of tall and slender cells

(2) Nucleus of the cell is located at its base

(3) Free surface may have microvilli

(4) It is commonly found in PCT of humans

159. Read the following statements A and B.
A: Compound epithelium has a limited role in secretion and absorption.
B: Compound epithelium covers the dry surface of the skin.
Choose the **correct** option.

(1) Both statements are incorrect

(2) Both statements are correct

(3) Statement A is correct but statement B is incorrect

(4) Statement B is correct but statement A is incorrect

160. The most abundant and widely distributed tissue present in humans is

(1) Epithelial tissue

(2) Connective tissue

(3) Muscular tissue

(4) Neural tissue

161. Additional volume of air that a person can expire by forcible expiration is known as

(1) Inspiratory reserve volume

(2) Expiratory reserve volume

(3) Functional residual volume

(4) Residual volume

162. Choose the **correct** option to complete the analogy
Respiratory rhythm centre : Medulla oblongata :: Pneumotaxic centre : _____

(1) Midbrain

(2) Medulla oblongata

(3) Pons

(4) Cerebellum

163. Read the given statements in context of humans.
(a) The lungs are situated in the thoracic chamber which is an anatomically air tight chamber.
(b) The anatomical setup of lungs in thorax is such that any change in the volume of the thoracic cavity will be reflected in the lung cavity.
Select the **correct** option.

(1) Only statement (a) is correct

(2) Only statement (b) is correct

(3) Both statements are correct

(4) Both statements are incorrect

164. In humans, the major part of carbon dioxide is transported

(1) As bicarbonate

(2) As carboxyhaemoglobin

(3) As carbamino haemoglobin

(4) In a dissolved state through blood plasma

165. Which of the following factors do **not** favour the formation of oxyhaemoglobin?

(1) High pCO_2 and low pH

(2) High pO_2 and high pH

(3) Lesser H^+ concentration and lower temperature

(4) Low pCO_2 and low H^+ concentration

166. Assertion : All the components of the nodal tissue are autoexcitable but only SA node acts as the pacemaker.

Reason : SA node can generate the maximum number of action potentials and is responsible for initiating and maintaining the rhythmic contractile activity of the heart.
In the light of above statements, select the correct option

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false

(4) Assertion is false but Reason is true

167. The state of heart when it is not pumping blood effectively enough to meet the needs of the body is called

(1) Heart attack

(2) Cardiac arrest

(3) Heart failure

(4) Atherosclerosis

168. Match the following w.r.t. leucocytes in human blood and their percentage in total WBCs. Choose the **correct** option.

Column I		Column II
a. Basophil	(i)	60-65 per cent
b. Eosinophil	(ii)	0.5-1 per cent
c. Monocyte	(iii)	2-3 per cent
d. Neutrophil	(iv)	6-8 per cent

(1) a(ii), b(i), c(iv), d(iii)

(2) a(iii), b(i), c(iv), d(ii)

(3) a(ii), b(iii), c(iv), d(i)

(4) a(ii), b(iii), c(i), d(iv)

169. The pulmonary circulation starts by the pumping of __A__ blood by the right ventricle which is carried to the __B__.

Choose the **correct** option that fills the blanks A and B respectively.

(1) Oxygenated; Heart

(2) Oxygenated; Lungs

(3) Deoxygenated; Lungs

(4) Deoxygenated; Heart

170. In humans, the structure **not** found in the renal pyramid is

(1) Convoluted tubules

(2) Collecting ducts

(3) Vasa recta

(4) Loop of Henle

171. Which one of the following does **not** constitute a part of uriniferous tubule?

(1) Distal convoluted tubule

(2) Collecting duct

(3) Bowman's capsule

(4) Loop of Henle

172. Read the following statements and select the **correct** option.

Statement A : Rh incompatibility between Rh –ve mother and Rh +ve foetus can lead *to erythroblastosis foetalis* in baby.

Statement B : AB^+ person can accept blood from AB^-, A^+, B^- and O^+ person.

(1) Both statements are correct

(2) Only statement A is correct

(3) Only statement B is correct

(4) Both statements are incorrect

173. In the kidneys, many of the straight tube called _____ converge and open into the renal pelvis through medullary pyramids in the calyces.

Choose the **correct** option that fills the blank.

(1) Proximal convoluted tubule

(2) Henle's loop

(3) Distal convoluted tubule

(4) Collecting duct

174. Correct order of increasing toxicity among nitrogenous wastes ammonia, urea and uric acid is

(1) Uric acid < urea < ammonia

(2) Uric acid < ammonia < urea

(3) Ammonia < urea < uric acid

(4) Urea < uric acid < ammonia

175. Match column I with column II and select the **correct** option:

Column I		Column II
(a) Ketonuria	(i)	Rapid spasm in muscles
(b) Tetany	(ii)	Insoluble mass of crystallised salts formed within the kidney
(c) Renal calculi	(iii)	Accumulation of urea in blood
(d) Uremia	(iv)	Presence of ketone bodies in urine

(1) a(ii), b(iii), c(i), d(iv)

(2) a(i), b(ii), c(iii), d(iv)

(3) a(iii), b(ii), c(iv), d(i)

(4) a(iv), b(i), c(ii), d(iii)

176. Total number of tarsals in both lower limbs of humans is equal to

(1) Total number of phalanges in both upper limbs

(2) Total number of phalanges in right hand

(3) Total number of bones in left upper limb

(4) Total number of vertebrae in vertebral column

177. Which type of joint is present between the atlas and axis vertebrae?

(1) Pivot joint

(2) Gliding joint

(3) Hinge joint

(4) Saddle joint

178. The thick filaments are held in the middle of the sarcomere by a thin membrane called __(i)____ located in the centre of ____(ii)____

Select the option which fills the blanks (i) and (ii) correctly

(1) (i) M - Line (ii) I - Band

(2) (i) Z - Line (ii) A - Band

(3) (i) Z - Line (ii) H - Zone

(4) (i) M - Line (ii) A - Band

179. Select the **incorrect** difference between red and white muscle fibres.

	Red muscle fibres	White muscle fibres
(1)	More number of mitochondria	Less number of mitochondria
(2)	More myoglobin	Less myoglobin
(3)	More sarcoplasmic reticulum	Less sarcoplasmic reticulum
(4)	Carry out aerobic oxidation	Depends on anaerobic process for energy

(1) (1)

(2) (2)

(3) (3)

(4) (4)

180. How many bones given in the box is/are included in axial skeleton of humans?

Lacrimal, Coxal, Scapula, Sternum, Patella

Choose the **correct** option.

(1) One

(2) Two

(3) Three

(4) Four

181. A part of brainstem which contains centres for controlling respiration, cardiovascular reflexes and gastric secretion is

(1) Midbrain

(2) Thalamus

(3) Medulla oblongata

(4) Hypothalamus

182. Read the following statements and choose the **correct** option

Statement A : Dendrites conduct nerve impulses towards the cell body or cyton

Statement B : From cyton, the axon conducts the nerve impulse towards the nerve endings

(1) Statement A is correct but statement B is incorrect

(2) Statement B is correct but statement A is incorrect

(3) Both statements are correct

(4) Both statements are incorrect

183. How many axons and dendrites are usually present in the multipolar neurons?

(1) One axon and one dendrite

(2) One axon and two or more dendrites

(3) Two axons and one dendrite

(4) Two dendrites and three axons

184. In a chemical synapse, the receptors for neurotransmitters are present on/in the

(1) Pre-synaptic membrane

(2) Synaptic cleft

(3) Post-synaptic membrane

(4) Axon

185. Four round swellings, called corpora quadrigemina are present in

(1) Forebrain

(2) Midbrain

(3) Hindbrain

(4) Spinal cord

SECTION-B

186. A attach skeletal muscles to bone whereas B attach one bone to another. Here A and B are

(1) A-Ligaments; B-Tendons

(2) A-Tendons; B-Ligaments

(3) A-Tendons; B-Adipose tissue

(4) A-Cartilage; B-Ligaments

187. The tissue which exerts greatest control over the body's responsiveness to stimuli is

(1) Cardiac muscle tissue

(2) Neural tissue

(3) Skeletal tissue

(4) Blood

188. Heterocyclic ring is not present in

(1) Uracil

(2) Cholesterol

(3) Adenylic acid

(4) Guanosine

189. In case of amino acid named serine, the variable group is

(1) $-CH_3$

(2) $-H$

(3) $-COOH$

(4) $-CH_2OH$

190. Number of carbon atoms in arachidonic acid is _____, whereas in palmitic acid is _____ excluding their carboxyl carbon.
Select the option that **correctly** fill the blanks respectively.

(1) 20, 16

(2) 16, 20

(3) 19, 15

(4) 15, 20

191. Choose the **incorrect** statement w.r.t. human's sound box.

(1) It is a cartilaginous structure

(2) It is also called larynx

(3) It is common passage for food and air

(4) Epiglottis prevents the entry of food into the larynx

192. Structure(s) **not** supported by incomplete cartilaginous rings is/are

(1) Initial bronchioles

(2) Tertiary bronchi

(3) Trachea

(4) Terminal bronchioles

193. The **correct** route through which impulse travels in the heart is

(1) AV node → Bundle of His → SA node → Purkinje fibres

(2) SA node → AV node → Bundle of His → Purkinje fibres

(3) AV node → Purkinje fibre → SA node → Bundle of His

(4) SA node → Bundle of His → AV node → Purkinje fibre

194. Which of the following blood cells are responsible for inflammatory reactions and secreting serotonin and histamine?

(1) Basophils

(2) Macrophages

(3) Monocytes

(4) Neutrophils

195. Other than kidneys, how many of the following organ(s) also help in the elimination of excretory wastes in humans?

Liver, Skin, Lungs, Sebaceous gland, Salivary gland

Select the **correct** option.

(1) Four

(2) Five

(3) Two

(4) One

196. Select the **incorrect** statement w.r.t. ultrafiltration.

(1) The glomerular capillary blood pressure causes filtration of blood through three layers.

(2) Podocytes are the epithelial cells of Bowman's capsule.

(3) Podocytes are arranged in an intricate manner so as to leave some minute spaces called slit pores.

(4) Blood is filtered so finely through the filtration membrane, that almost all the constituents of the blood except amino acids pass onto the lumen of the Bowman's capsule.

197. The non contractile, regulatory proteins helpful in muscle contraction are

(1) Actin and tropomyosin

(2) Tropomyosin and myosin

(3) Actin and troponin

(4) Tropomyosin and troponin

198. Which of the following will remain unchanged during contraction of skeletal muscle fibres?

(1) Length between two consecutive Z-lines

(2) Length of actin filaments

(3) Length of an isotropic band

(4) Length of H-zone

199. Depolarisation of neural membrane is caused due to

(1) Rapid efflux of Na^+

(2) Rapid influx of K^+

(3) Rapid influx of Na^+

(4) Rapid efflux of K^+

200. Arrange the following cranial meninges in correct sequence from outer to inner side.
A – Pia mater
B – Arachnoid
C – Dura mater

(1) A – B – C

(2) B – A – C

(3) C – B – A

(4) B – C – A